PV102 Day 1: Applications and Design Decisions

Course transition from verified workmanship to system-level judgment

CTS Day 17	10:00 AM-3:00 PM	6 students

Instructor Run Sheet | Audience: Instructor or substitute | Monday, August 3, 2026

Core rule

Start with load, site, and need. PV101 proved that students can assemble and verify a small system. PV102 begins by deciding which system belongs in the job.

Learning targets

- Distinguish grid-tied, hybrid, off-grid, and PV-direct architectures.
- Trace a first-pass energy path from PV source to the intended load.
- Identify load, site, access, budget, maintenance, and resilience constraints.
- Defend an architecture choice without guessing beyond the available evidence.
- State protection, isolation, and verification boundaries before construction.

Readiness and materials

Before students arrive	Materials
• Open slide deck and test display. • Print six quizzes, one student packet per student, and one lab scenario per team. • Set three team stations with chart paper, markers, calculators, and component cards. • Cover or isolate PV modules; keep batteries and electrical hardware disconnected.	• Projector and slide deck • Printed quizzes and packets • Three component-card sets or de-energized components • Chart paper, markers, calculators, clipboards • PPE only if students handle physical components

Five-hour run of show

Time	Block	Instructor move
10:00-10:15	**PV102 transition**	Name the progression: verified workmanship now supports design judgment. Introduce the core rule.
10:15-10:40	**Morning quiz and review**	Students work independently for 12-15 minutes. Review by asking for evidence, not just letters.
10:40-11:30	**Applications lesson**	Use slides 1-16. Pause after each architecture and ask what client need it serves.
11:30-12:15	**Guided architecture sort**	Read three client prompts aloud. Students choose an architecture, then revise when a new constraint is revealed.
12:15-12:45	**Lunch**	Reset three lab stations. Keep all electrical equipment disconnected.
12:45-1:00	**Lab brief**	Assign teams and roles. Model the acceptance criteria and the two-minute handoff.
1:00-2:15	**Scenario design lab**	Teams complete one primary scenario. Rotate if finished; instructor probes assumptions and safety boundaries.
2:15-2:40	**Handoffs and peer verification**	Each team gives a two-minute handoff. Peers check need, path, constraint, boundary, and unknowns.
2:40-3:00	**Exit ticket and reset**	Collect packets. Preview Day 2: site and solar-resource assessment.

Talk track: course transition

Opening script

"You already know how to connect, inspect, measure, and hand off a small PV system. PV102 asks a harder question: what system should exist in the first place? A professional starts with the client's load, the site, and the real operating need."

Architecture teaching notes

Architecture	Ask students	Listen for
Grid-tied	What job does the utility perform?	Normal grid support; bill reduction; no assumed outage operation
Hybrid	Which exact loads need backup, and for how long?	Defined resilience goal; storage has a specific job
Off-grid	What happens through the night or poor weather?	Load continuity, autonomy, service access, and storage
PV-direct	Can the load wait for sun?	Flexible timing or process storage such as a water tank

Guided architecture sort

1. Start with: "The client wants solar." Reject this as incomplete. Ask: what must the system accomplish?
2. Reveal a utility connection. Students choose between grid-tied and hybrid only after the outage requirement is stated.
3. Reveal a remote site. Ask whether the load must operate at night; use that answer to distinguish off-grid from PV-direct.
4. For every answer, require one sentence of evidence and one unknown that still needs verification.

Lab coaching prompts

- What must run, and when must it run?
- What does the site allow or prevent?
- What job would a battery perform here?
- Where must energy be isolated or overcurrent limited?
- Which statement is verified, and which is still an assumption?
- What would make you change this architecture choice?

Common misconceptions and corrections

Misconception	Correction
Every solar system needs batteries.	Storage is selected only when it performs a defined job.
Grid-tied solar automatically powers a building during an outage.	Outage operation requires equipment and design specifically intended for that function.
The biggest array is the best design.	The best design matches the load, site, goals, and constraints.
Protection can be added after the layout is complete.	Protection, isolation, labeling, and access belong in the first design pass.
A missing fact can be replaced with a reasonable guess.	Record the unknown and identify how it will be verified.

Assessment evidence

- Quiz: prerequisite electrical and verification habits.
- Design brief: architecture choice, energy path, constraints, boundaries, and unknowns.
- Handoff: scope, decision, evidence, safety boundary, and unresolved items.
- Exit ticket: operating need, storage decision, and Day 2 verification question.

Minimum successful performance

The student selects a defensible architecture, maps a plausible major-component path, cites scenario evidence, names a protection or isolation boundary, and records unknowns instead of guessing.

Answer snapshot and contingency plan

Scenario anchors

- Remote trail counter: off-grid; day-and-night operation and no utility support make storage and serviceability central.
- Small shop: hybrid; grid-connected bill reduction plus a defined limited-backup job.
- Daytime water pump: PV-direct is the strongest first-pass choice; the tank provides process storage and night operation is unnecessary.

If equipment or printing is unavailable

- Run the entire lab with handwritten component cards. No electrical hardware is required.
- Project the scenarios and have teams build one design brief on chart paper.
- If the slide deck cannot open, use the Whiteboard Outline and the architecture reference table.
- If time is short, complete one scenario deeply and use the other two as oral checks.

Early-finisher extension

- Add one new client constraint that forces a redesign.
- Compare a no-storage option with a storage option and name the job the battery performs.
- Create a verification plan for every unknown in the brief.

Safety and scope boundaries

No energized work

This lesson does not authorize live electrical work, utility interaction, battery connection, PV module connection, or code-compliance approval. Use covered/isolated modules and disconnected components only. Follow manufacturer instructions and the program's established electrical safety procedures in later hands-on work.

Day 2 handoff

Carry forward the three design briefs and their unknowns. Day 2 will convert vague site assumptions into observations: solar access, shade, orientation, space, weather, access, and maintenance conditions.

Instructional sources

- U.S. Department of Energy, Solar Energy: https://www.energy.gov/topics/solar-energy
- U.S. Energy Information Administration, Solar explained: https://www.eia.gov/energyexplained/solar/
- OSHA, Green Job Hazards - Solar Energy: Electrical: https://www.osha.gov/green-jobs/solar/electrical
- OSHA, Green Job Hazards - Solar Energy: Lockout/Tagout: https://www.osha.gov/green-jobs/solar/lockout-tagout
- Sean White, Solar Photovoltaic Basics, 2nd ed., Routledge, 2018, ISBN 9781138102866.